BUILD MASTER GUIDE

For VEX Robotics. V5. Made in May 2023.

Disclaimer: this is a study guide for all, so it will contain a mix of simple and more complex information. Ideally, a builder should already know most of this information and this should be a review for the competitive exam (or something else), but if not, learn ahead!

Please note that hyperlinks are added throughout this document. If you are interested in learning more about the topic, feel free to click them. This document contains content from VEX slideshows, novice lessons, the Purdue Sigbots wiki, and other online resources.

Table of Contents

**the important and slightly more complicated concepts are highlighted in the table of contents below

Subsystem Specific 3

Drivetrain 3

- Tank (commonly used with 4 or 6 motors) 3
- Mecanum 3
- X-Drive/Holonomic 3
- H-Drive (one of the most inefficient drives that exist) 3

Good Drivetrain Building Practices 4

Intake/Manipulators 5

- Conveyor (chain & flaps) 5
- Flex wheels 5
- Claw (can be powered by pneumatics or motors) 5
- Scoop 5
- Plow (passive) 5
- Piston Powered 5
  - Clamp 5
  - Indexer 5

Things to consider for intakes 6

Shooters 6

- Flywheel 6

- Catapult 6
- Linear Puncher 6

Lift 7

- Bar lifts: 2, 4, 6, etc. bar 7
- Chain bar 7
- Reverse double 4-Bar (RD4B) 7
- Scissor 7
- Cascade/chain 7
- Peaucellier 7

Build Quality 8

Types of Bracing 8

Boxing 8

Friction/Slop 9

Shoulder screws 9

High strength (HS) gears and sprockets 9

Sprockets and Chain 9

• When chaining, avoid using 6T sprockets (especially when paired with larger size sprockets) since usually only 2-3 are in contact with the chain, making it prone to slipping 9

Joints 10

Axle joints 10

Screw joints 10

Power transfer 11

Power sharing 11

Pneumatics 13

Mechanical Advantage 13

Other Resources 13

Subsystem Specific

Drivetrain

Note: always keep the drivetrain as close to a square as possible; the more you deviate from a square, the more torque the drive will require to turn

- Tank (commonly used with 4 or 6 motors)

Pros	Cons
Keeps a lot of open space on the drivetrain	Can't strafe
Easy to build and program	Can be pushed from side when it doesn't have middle locked omnis or flex wheels
Most amount of torque	

- Mecanum
  - Same as a tank drivetrain except that it uses mecanum wheels

Pros	Cons
Good amount of open space (less than a tank drive)	14% less torque and 6% less speed due to angle of rollers (values experimentally derived)
Can strafe	
Hard to push from side	Strains motos and axles when strafing

- X-Drive/Holonomic

Pros	Cons
$\sqrt{2}$ times faster than other drives (fastest)	$\sqrt{2}/2$ times torque (weaker)
Most accurate at turns	Less amount of open space on drive
Most efficient at strafing	

- H-Drive (one of the most inefficient drives that exist)

Pros	Cons
Has a strafing ability	Takes an extra motor and more space for strafing
	Strafing is extremely weak and slows down driving/turning

- Other types of drives: kiwi (basically x-drive with 3 wheels/motors), track drive (tread instead of wheels)

Good Drivetrain Building Practices

- Out of all subsystems, having the least friction and slop is most important on the drivetrain
    - It carries the weight of the entire bot and overheats (the more friction the drive has, the more energy/force it will take to move the drive, the more easily the motors will overheat/ battery will die)

Intake/Manipulators

Types of Intakes:

- Conveyor (chain & flaps)
    - Great for moving a relatively light object over a fairly long distance
    - Can also be used around a sprocket with long flaps as an intake
    - Con: can break if used with heavy loads or if it has too much tension (not enough tread/chain)
- Flex wheels
    - This is basically the same as taking a wheel, adding rubber bands, and then using that for your intake, except that flex wheels compress more and can come in smaller sizes
    - If you want to transfer the game element over a larger distance (be it higher and/or farther), you will likely need several stages of the intake
- Claw (can be powered by pneumatics or motors)
    - Holds onto game elements
    - Rubber bands or mesh can be added for extra grip
- Scoop
    - Uses the momentum of the robot to pick up game elements
- Plow (passive)
    - It is like a bulldozer, in the sense that it collects and pushes away game elements (usually spheres or some other object that can easily slide across the field tiles)
    - It is best to have a concave plow to ensure that you can push away as many game elements as possible
        - Alternatively, you can have a flat plow with guards on the 2 ends, making a rectangle with an open side
- Piston Powered
    - Clamp
        - Applies the concept of a basic lever in which an upwards force from the piston (mounted on the bottom of the lever) pushes the other end of the lever down which clamps/holds onto the game object
        - The far end of the lever usually has standoffs/c-channels and/or mesh to grip the game object
    - Indexer
        - Shooting indexer: the piston can directly push the object to be shot to the flywheel or it can push a plate/piece of polycarb that acts as the lever

Things to Consider for Intakes:

- Margin of error
    - You want a wider intaking range so that the game elements can be intake with minimal alignment work
    - Funnels help with this, but pushing the game element towards the intake as the robot drives forward
- Speed (usually, the higher the better, but it also must have enough torque)
- Capacity: how many game elements must you intake at a time
    - Higher capacity = higher torque = lower speed
- Distance covered
    - How high and far into the robot does the game object need to go?
    - How many stages will you have to ensure this?
- Just remember that what looks good on paper may not actually be the best for your bot in real life, so do lots of testing to see how game objects interact with your intake
- Remember to be *creative* with your designs

<u>Shooters</u>

- Flywheel
    - Great for back-to-back shooting

Pros	Cons
Extremely low rebound time	Requires an indexer in most cases
	High startup time and motor may overheat over time

- Catapult
    - Great for when you want to shoot 1 or only a few things at a time

Pros	Cons
Does not require an indexer	Pretty high rebound time + need a slip gear
Great for some niche circumstances	Damages motor over time

Tuning a catapult

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- Linear Puncher

Pros	Cons
Fairly low reload time	May damage the gears over time

<u>Lift</u>

- Bar lifts: 2, 4, 6, etc. bar
    - 4+ bars keep the bar on the far side of the lift parallel as the lift tower (good for when game objects need to be lifted to moderately high heights and then scored)
    - Anything over an 8 bar usually isn't practical to implement
- Chain bar
    - Like a 4-bar, but works by 2 sprockets (on ends) + chain keeping the bars parallel
- Reverse double 4-Bar (RD4B)
    - It is really good when you need to go really high and want the lift to be compact when folded

Pros	Cons
Can go really high	Really heavy; shouldn't be used when a 4 bar can work
Has a good center of mass when extended	Difficult to make for first time / new builders
Strong when braced correctly	Can self destruct when braced incorrectly

- Scissor
    - Motor turns a gear that translates along a rack gear, which moves the bottom scissor bar of the lift
        - Requires a high torque ratio since the lift is extremely heavy
    - Repeated scissor mechanisms (basically an x with a joint in the middle) connected to each other
    - Needs to have 2 braced, side-to-side scissor mechanisms for the lift to be structurally sound
    - Fairly compact when folded (height-wise) but can go really high up when lifted
- Cascade/chain
    - Essentially multiple stages of elevators which are connected to each other and can fold to take up very little space
- Telescoping
    - Same as chain bar, except it uses rope instead of chain
- Linear slide
    - A regular gear goes up a rack gear with some sort of system mounted on it

    - Hardly used
- Elevator
    - Essentially a vertical conveyor with a bucket (or similar mechanism) that allows
- Peaucellier
    - Essentially a 4 bar with other linkages that allow the system mount on the far end of the linkage to go up and down in a straight line, parallel to the lift tower

Build Quality

Types of Bracing

- Cross
    - Perpendicular piece of metal across 2 parallel pieces of metal (usually c channel)
    - Generally used for the drivetrain and connecting 2 sides of a lift
    - Most common
- X-bracing
    - Generally used for bracing 2 sides of a lift
    - The X has triangles inside of it, which is the strongest shape, making the bracing strong
    - Can be done with c-channels or half cuts (pretty much any metal other than standoffs)
    - Heaviest type of bracing because it has 2 channels instead of 1
        - Note: the 2 channels of the "X" should be braced together with a screw at the center of the X
- Triangle bracing
    - Usually used for bracing towers
    - One of the best forms of bracing in general because triangles are the strongest shape
    - Use the pythagorean theorem to determine the number of holes of the c-channel and its placement (this is where knowing your pythagorean triples is important!! Or just google it)
        - ex. If you have 5 holes worth of space horizontally, the tower goes 5 segs high (25 holes), and you don't care about how length of the half cut that will be used for bracing, then place the half cut 5 holes away from the tower horizontally, 12 holes vertically, and make the half cut 13 holes long → (5, 12, 13) is a pythagorean triple)
    - The strongest type of triangle is an equilateral triangle (all side lengths and angles are the same), but it isn't necessary possible to use this in every scenario
        - ex. This wouldn't be possible to implement for tower triangle bracing with half cuts because it wouldn't

Boxing

- It is a good idea to box brace any metal that is under a lot of force
- FOR C-CHANNELS: Done by adding a 1/2" and 3/8" spacer in the middle of the c-channel, and holding it with a screw and locknut
    - Alternatively can be done with a combination of standoffs (i.e. 3/4" + 1/8")
- To box brace a u-channel, use a 1" spacer
- Half cuts cannot be boxed
- 3 and 5-wides can be boxed, but they are hardly used and even more rarely boxed

Friction/Slop

- For all joints, it is important to maintain low levels of friction and slop
    - The two are inversely related - if you increase slop, the decrease friction
    - At a certain point, if you have a crazy large amount of slop, the lateral movement of the mechanism that is "part" of the joint will add to friction instead of removing from it
- To decrease friction, always choose blue cartridges (600 RPM) whenever possible
    - Red (100 RPM) and green (200 RPM) cartridges have 2 stages of gears (compound gear ratio) while blue (600 RPM) only has 1
    - Therefore blue has less friction (pretty noticeable but not too significant)
        - Source: Sharwin, VEX Forum
- The main purpose of bearings is to reduce friction
    - It works because
        1. Axles/screws (metal) hit plastic instead of metal
        2. Axles rotate inside a circular shape (bearing hole, rather than a square one (c-channel hole)
    - Make sure they are perfectly mounted onto the c-channel
        - Use shoulder screws
        - The square 'guides' help keep the bearing aligned to the c-channel hole
        - The main goal is to keep the axle from hitting the c-channel at all
            - Can be tested by mounting the bearing, putting an axle through it, and listening for an metal sound (it is higher pitched)
    - Bearings are also used in screw joints for the same reason as above, except that a screw is going through the bearing instead of an axle

Shoulder screws

- These screws have a wider, non-threaded region near the head such that it fits in a c-channel with nearly no slop/wobble
- Most friction problems for joints arise from the misalignment of c-channels (i.e. on a drivetrain, is the bracing c-channels are misaligned then there might be more friction on the wheel/gear joints)
    - This screw fixes that
- Great for anything that needs better alignment
    - Drive chassis
    - All bearings

High strength (HS) gears and sprockets

- Almost always use them
- Pros: gears don't slip on one another, can withstand a lot of force (gets damaged less easily), can be used with both LS and HS gears
- Cons: more surface area ⇒ more friction, take up more space

Sprockets and Chain

- When chaining, avoid using 6T sprockets (especially when paired with larger size sprockets) since usually only 2-3 are in contact with the chain, making it prone to slipping
- Tuning the tightness of chain:
    - It the chain does not move at all when you push it or it feel really tense, that is an issue
        - Add at least 1 chain link
    - If the chain moves more than 2 centimeters then remove a chain link
    - If you can't find the perfect balance between chain tension, keep the extra link and add a tensioner
        - This is bunch of free spinning spacers that push out the chain to tension it a bit
        - You can tune how much tension the chain has by moving the mounting point for the spacers

Other Build Quality Tips

- Never let metal contact metal (ex. c-channels on c-channels, axles on c-channels)
    - Use bearings, spacers, etc. depending on the situation

Joints

Axle joints

- Required to be used when a joint is connected to a motor

Pros	Cons
Relatively simple to make	Bulky and heavy

Screw joints

- Commonly used for all joints other than those that have a motor

Pros	Cons
Low slop and friction	May be confusing to build a first, but really easy once you understand
Light and compact	

- How to build a screw joint and how it works:
    - Put a screw through a hole on the rigid c-channel (usually the base of a drivetrain or a tower)
    - Secure it with a keps nut (the keps nut should be fully tightened)
        - This will hold the screw in place and prevent it from wiggling or rotating
    - Add a nylon spacer (to prevent metal on metal contact)
    - Mount a bearing onto the moving channel and add it onto the screw
    - Add another nylon spacer (to prevent metal on metal contact)
    - If cantilevered (meaning that the joint is only secured on one side, i.e. 4-bar arm is secured to only 1 lift tower; usually bad because the screw/axle is prone to bending, which will increase friction)
        - Add the locknut and tighten//loosen to tune the screw joint
    - If not cantilevered (the joint is held by 2 rigid c-channel and holds a moving channel
        - Fill in the space betweening the moving channel and 2nd rigid channel with spacers (leave between a 1/32" and 1/16" of spacing for the screw joint to be able to move freely)
        - Add a bearing on the 2nd rigid c-channel
            - This is so that the screw can't bend up or down since the channel holes are so big
            - It is <u>not</u> to reduce friction since the screw is <u>not</u> rotating
        - Tighten the screw such that it goes through the 2nd rigid channel

        - Add a locknut and tighten/loosen to tune

Power transfer

- Gear
    - Gear ratios
    - Compound gear ratios
- Chain

Power sharing

- Ratchets
    - How they work
        -
    - How to build (2 main types)
        - Ratchet is 'inside' gear (most common): https://youtu.be/rrg5N7xTczM
            - .
        -
    - Using ratchets to power share
        - Possibly OP Motor Sharing Mechanism
        - 315G's Turning Point Ratchets Explained
        - 6671X Single Motor Intake + Indexer
- Differential
    - ITZ Drive-Four Bar differential
- PTO (power take-off)
    - This is when you want to make motor(s) away from a subsystem to temporarily power another
    - You can switch between which system receives these shared motor(s)
    - Most common with the drive and another subsystem (ex. intake, lift)
        - Have a standard 4 motor drive
        - 2 more motors can switch between powering the drive or the other subsystem by using a piston to move gears such that the motor power the other subsystem
    - How to implement (2 different ways)
        1. 606X Spin Up | Drivetrain PTO Explanation
        2. Motors on a pivot, switching between powering the drive and intake
            - VEX Simple Power Take-Off (PTO)/Transmission Tutorial
        - Other
            - 606X Spin Up | Flywheel PTO Explanation / Ben Face Reveal

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Pneumatics

- Reservoirs: tanks (up to 2 allowed) that store pressurized air up to a 100 PSI
- Tubing: (usually black) polyurethane tube that is used to transfer air throughout the pneumatic system, connecting reservoirs, with solenoids, with pistons
- T-Connector: a part which can split air from 1 tube into 2 different tubes; has 1 'input' and 2 'output'
- Solenoid
- Piston: a rod that extends outwards when pressurized air is pushed through the solenoid and into the piston
    - Single acting: has a passive, spring retraction system that does *not* use air
    - Double acting: air can flow both ways, meaning it can extend and retract with the force of air
- How to set up a full pneumatic system
    - Insert picture
- How to optimize/properly set up a pneumatic system
    - Imagine you want to extend and then retract a double acting piston. Firstly, all the air in the piston will enter from the bottom valve to extend the piston. Then, that air will be dumped and air will be sent to the top of the piston to retract it. Therefore, any air that is either in the piston or in the tubing before the solenoid will be dumped when switching the status of the piston between extended/retracted. So, do the following to optimize (save) air in your pneumatic system:
        - Maximize tubing before solenoid(s)
        - Minimize tubing air after solenoid(s)
    -

Mechanical Advantage

3 types:

- Force
- Distance/Displacement
- Speed

Other Resources

- BLRS Wiki
- VEX FORUM
    - Had a lot of helpful information
    - If you have a question that isn't already there, ASK!
- https://vexwiki.org/

    - There isn't much content on this website, but something checking out if interested
- Wiki - VEX V5 General Discussion - VEX Forum
    - This forum thread has links to many resources
    - Contains a lot of valuable information and definitely worth checking out
- VEX Curriculum
- VEX ROBOTICS COMPETITION
- Robotics Reference Guide
- Drivetrains & Wheels :: Bionics Docs